Gold Future/ETF Analysis

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2025

1 Spot and Futures Analysis

It should be noted that all analyses, apart from those with train versus test groups, are performed on the full dataset available. Often this differs between analyses due to data availability and differing sizes in the intersections of different datasets.

Maturity	**Slope**	**Intercept**	R^2	**RMSE**
1-Month	0.99962	1.2811×10^{-7}	0.94937	0.00221
2-Month	1.00241	-1.3769×10^{-6}	0.94710	0.00227
6-Month	1.00322	8.1922×10^{-7}	0.94068	0.00242
12-Month	0.94779	2.2525×10^{-5}	0.83986	0.00409

Table 1: New Data: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

The slopes for the 1, 2, and 6-month contracts are all very close to 1, indicating a strong correlation with spot gold returns. The 12-month contract, however, has a noticeably lower slope of 0.94779 and a significantly lower R^2 of 0.83986. This may be a result of the lower liquidity and trading volume of the longer-dated contract, making it less sensitive to daily spot price movements. The great difference in all statistics, though, could also indicate that the data for the 12-month contract data could be corrupted, recorded incorrectly, or otherwise flawed.

The slopes are also very close to 1, but consistently lower than the newer data. The R^2 values are also significantly lower than the newer data. This may be due to the smaller sample size of the older data and lower liquidity. However, the much greater consistency of the Leung & Ward slopes and the lower R^2 values could also indicate a procedural difference in how the statistics were calculated.

The slopes and R^2 values generally increase with longer holding periods but decrease with longer-dated contracts. This suggests that futures returns are

Maturity	Slope	Intercept	R^2	RMSE
1-Month	0.94314	2.4191×10^{-5}	0.80947	0.00530
2-Month	0.94301	1.0000×10^{-5}	0.80911	0.00530
6-Month	0.94348	3.1797×10^{-5}	0.80934	0.00530
12-Month	0.94358	-5.2333×10^{-5}	0.80984	0.00529

Table 2: Leung & Ward (2015) Data: Regression coefficients for one-day futures returns on one-day spot gold returns.

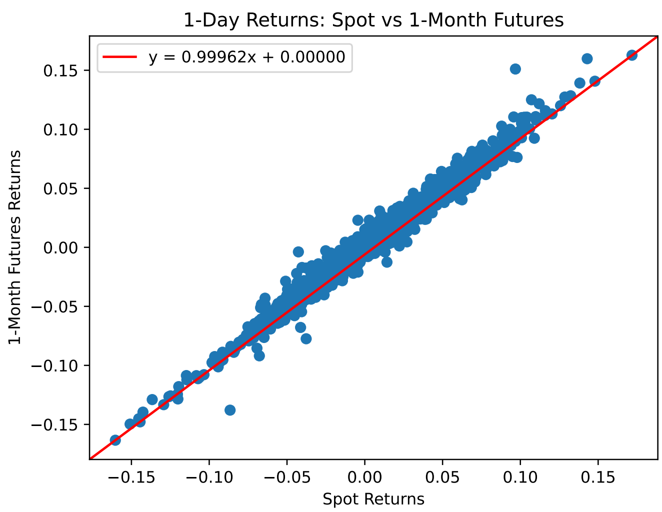


Figure 1: Scatter plot of one-day returns for spot gold versus 1-month gold futures.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.99962	1.00241	1.00322	0.94779
	5	1.00741	1.01021	1.01058	0.98753
	10	1.00224	1.00399	1.00337	0.98967
	15	1.00312	1.00494	1.00331	0.99205
R^2	1	0.94937	0.94710	0.94068	0.83986
	5	0.98419	0.98398	0.98117	0.95670
	10	0.99152	0.99101	0.98776	0.97473
	15	0.99386	0.99330	0.98994	0.98068

Table 3: New Data: Slopes and R^2 from the regressions of futures returns versus gold returns over different holding periods.

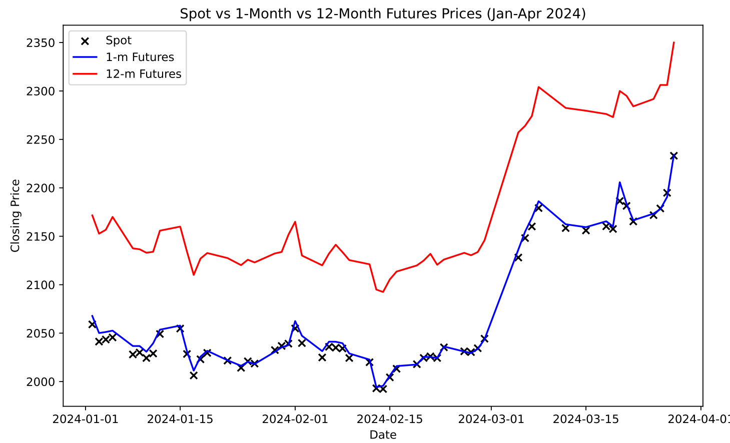


Figure 2: Comparison of closing prices for spot gold, 1-month futures, and 12-month futures in early 2024. The 1-month contract tracks the spot price closely, while the 12-month contract maintains a significant premium.

correlated with spot returns over longer horizons, with shorter measurement periods capturing more noise.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.94314	0.94301	0.94348	0.94358
	5	0.99805	0.99752	0.99757	0.99715
	10	1.01075	1.01020	1.00970	1.00841
	15	0.99448	0.99461	0.99431	0.99345
R^2	1	0.80947	0.80911	0.80934	0.80984
	5	0.97154	0.97158	0.97194	0.97165
	10	0.98516	0.98530	0.98572	0.98550
	15	0.98704	0.98691	0.98725	0.98713

Table 4: Leung & Ward (2015) Data: Slopes and R^2 from regressions of futures returns vs. gold returns over different holding periods.

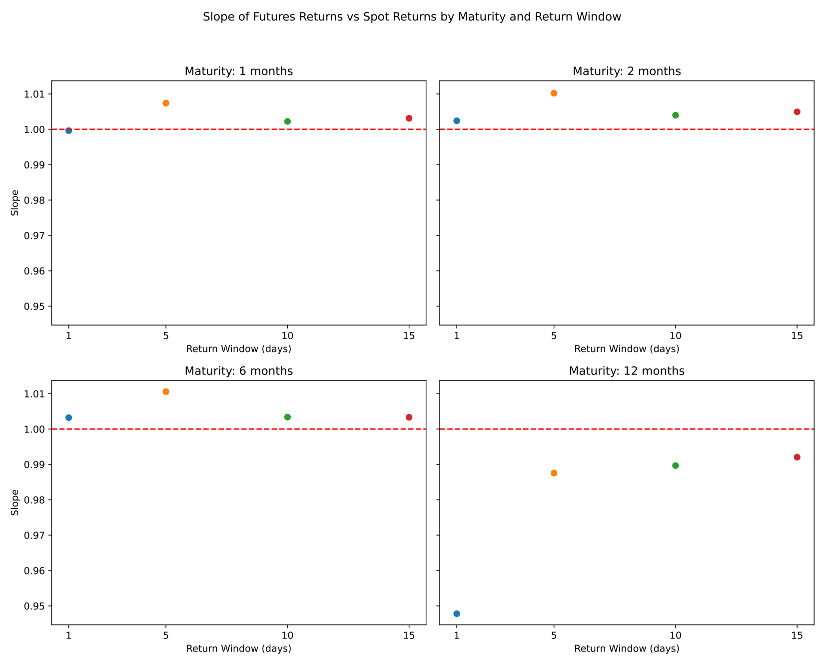


Figure 3: Regression slope of futures vs. spot returns, analyzed by contract maturity and the return window length in days. For most maturities, the slope approaches 1 as the return window lengthens, but the 12-month contract is still significantly different from 1 compared to the other maturities.

2 Spot Static Replication

The out-of-sample period is defined as starting on January 1, 2024 and is approximately 10% of the data. The RMSEs are calculated in dollar terms assuming a notional investment of $1000 in the spot gold price. The weights are calculated using a constrained least squares regression that forces the weights to sum to one.

Portfolio	w_0	w_1	w_2	w_6	w_{12}	$\mathbf{RMSE_{in}}$	$\mathbf{RMSE_{out}}$
1-m	0.00258	0.99741	—	—	—	$2.67	$5.12
2-m	0.00679	—	0.99322	—	—	$3.70	$5.69
6-m	0.02215	—	—	0.97785	—	$10.07	$13.34
12-m	0.04517	—	—	—	0.95483	$18.80	$17.72
1-m, 2-m	0.00170	1.21350	-0.21521	—	—	$2.62	$5.03
1-m, 6-m	0.00168	1.04713	—	-0.04881	—	$2.63	$4.86
1-m, 12-m	0.00192	1.01384	—	—	-0.01576	$2.68	$4.91
2-m, 6-m	0.00333	—	1.23020	-0.23352	—	$3.21	$4.62
2-m, 12-m	0.00311	—	1.09427	—	-0.09738	$3.29	$4.94
6-m, 12-m	0.00444	—	—	1.76141	-0.76585	$6.08	$11.87
1-m, 2-m, 6-m	0.00159	1.14787	-0.12240	-0.02706	—	$2.62	$4.93
1-m, 2-m, 12-m	0.00173	1.24621	-0.25302	—	0.00508	$2.66	$4.96
1-m, 6-m, 12-m	0.00201	1.10693	—	-0.17570	0.06676	$2.64	$4.52
2-m, 6-m, 12-m	0.00329	—	1.22206	-0.22099	-0.00436	$3.24	$4.43
1-m, 2-m, 6-m, 12-m	0.00190	1.22259	-0.14302	-0.14612	0.06465	$2.63	$4.60

Table 5: New Data: Weights and in/out-of-sample RMSEs for portfolios of 1, 2, 3, and 4 futures contracts.

The results show that using a single futures contract provides a reasonable replication, but performance diminishes as the maturity of the contract increases (out-of-sample RMSE rises from $5.12 to $17.72). Adding a second futures contract generally improves the out-of-sample RMSE, with the 2-month and 6-month combination performing best at $4.62. The best-performing portfolio overall is the three-contract combination of 2-month, 6-month, and 12-month futures, which achieved an out-of-sample RMSE of $4.43. Notably, several of these futures portfolios outperform the GLD ETF, which has an out-of-sample RMSE of $6.00 over the same period.

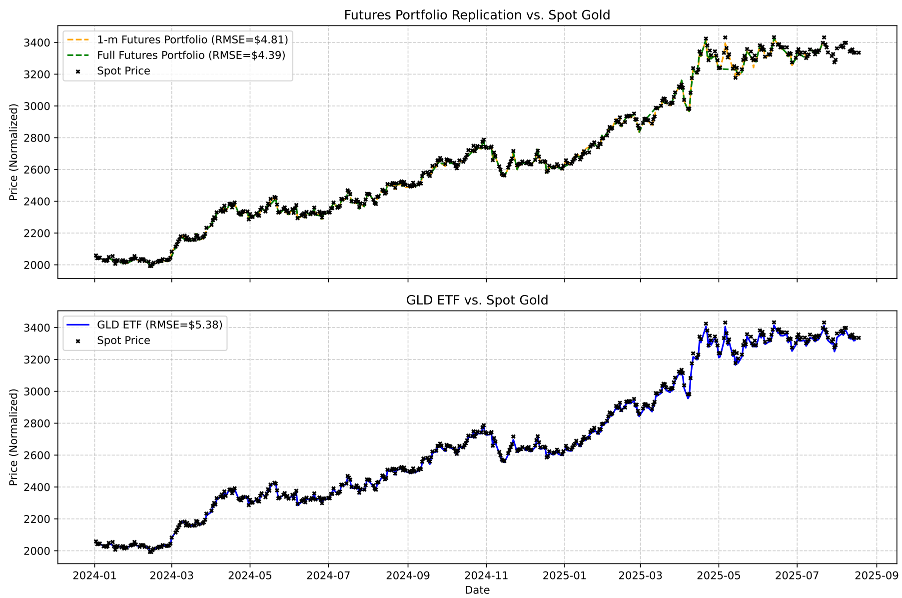


Figure 4: Visual comparison of tracking performance in early 2024. The top panel shows replication of spot gold using futures portfolios, while the bottom panel shows the GLD ETF's tracking of spot gold. The futures portfolios achieve a lower RMSE in this period than GLD.

Portfolio	w_0	w_{short}	w_{long}	$\mathbf{RMSE_{in}}$	$\mathbf{RMSE_{out}}$
1-m	-0.01071	1.01071	—	$6.63	$3.34
2-m	-0.04835	1.04835	—	$12.22	$6.12
12-m	-0.06842	1.06842	—	$15.11	$5.71
1-m, 2-m	-0.00088	1.27315	-0.27227	$6.27	$2.79
1-m, 6-m	-0.00171	1.23899	-0.23729	$6.27	$2.98
1-m, 12-m	-0.00021	1.19336	-0.19315	$6.29	$3.00
2-m, 6-m	-0.04079	5.06602	-4.02523	$10.27	$9.37

Table 6: Leung & Ward (2015) Data: Weights and in/out-of-sample RMSEs for portfolios of 1 and 2 futures contracts.

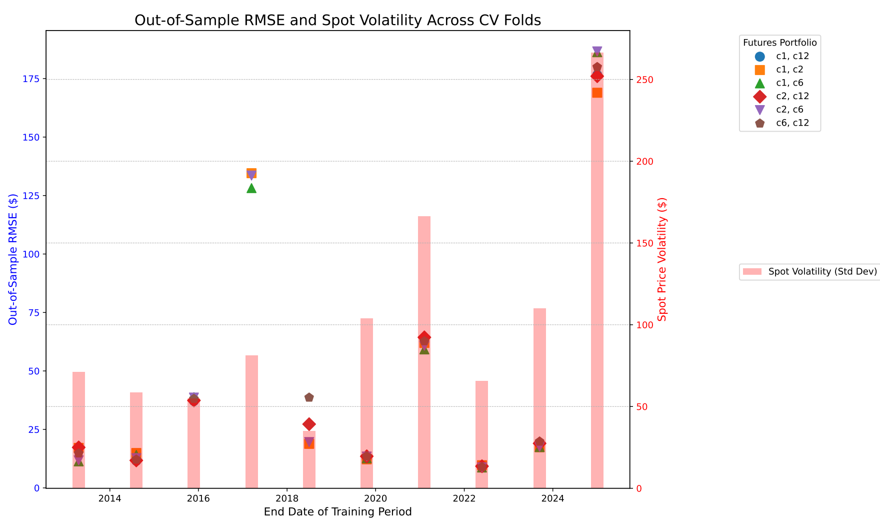


Figure 5: Out-of-sample RMSE for various futures portfolios across different cross-validation training periods. The pink bars represent the spot price volatility during each out-of-sample period.

3 Leveraged ETFs analysis

(L)ETF	Leverage (β)	Slope	Intercept	t-stat	p-value	R^2	RMSE
GLD	1	1.00431	-1.30×10^{-5}	2.039	0.042	0.9780	0.00164
UGL	2	2.01860	-2.10×10^{-4}	2.615	0.009	0.9688	0.00326
GLL	-2	-2.01174	1.96×10^{-4}	-1.663	0.097	0.9690	0.00323
UGLDF	3	3.00291	4.25×10^{-4}	0.026	0.979	0.2501	0.04519

Table 7: New Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

Table 8: Daily Return Differential Between LETF and Spot (Bias) Analysis: H_0 : Mean Bias = 0

ETF	Mean Daily Bias	t-stat	p-value
GLD	-0.000011	-0.5668	0.570828
UGL	-0.000202	-5.0935	0.000000
GLL	0.000190	4.8491	0.000001
UGLDF	0.000426	0.7767	0.437311

The regression results indicate that GLD, UGL, and GLL are effective at tracking their underlying benchmarks, with high R^2 values and slopes very close to their target betas. However, the low p-values for GLD (0.042) and UGL (0.009) suggest that their slopes are statistically different from the target betas of 1 and -1, respectively. UGLDF performs poorly, with very low R^2 values (0.2501), indicating they do not reliably track their leveraged benchmarks. The daily return differential analysis shows that UGL and GLL have statistically significant biases, while GLD and UGLDF do not.

(L)ETF	Leverage (β)	Slope	Intercept	t-stat	p-value	R^2	RMSE
GLD	1	1.00540	-1.6427×10^{-6}	1.42692	0.15382	0.98060	0.00163
UGL	2	2.00572	-1.3107×10^{-4}	0.73319	0.46351	0.96257	0.00350
GLL	-2	-2.00609	1.2541×10^{-4}	-0.76019	0.44723	0.96317	0.00346
UGLDF	3	3.01189	-1.9048×10^{-4}	0.11181	0.91099	0.28014	0.04369

Table 9: Leung & Ward (2015) Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

4 Leveraged ETFs Static Replication

Table below presents the optimal weights and RMSE metrics for static replication of leveraged positions using single futures contracts. The the future maturities that weren't used were not able to converge to a solution. Clearly, static replication of leveraged ETFs is ineffective and infeasible, and even moreso for inverse leveraged ETFs. The large out-of-sample RMSE values could also be attributed to the high volatility in the out-of-sample period. The general ineffectiveness does make sense given the path-dependent nature of leveraged ETFs, which cannot be captured by a static portfolio.

Table 10: Static Replication for Leveraged Targets: Optimal Weights and RMSE

Target Leverage	**Contract**	$w_{\mathbf{cash}}$	$w_{\mathbf{future}}$	**RMSE (In)**	**RMSE (Out)**
-3x (DGLD)	c1	2.75	-1.75	$89.90	$9,096.48
	c2	2.75	-1.75	$89.56	$9,814.67
	c6	2.72	-1.72	$90.25	$16,591.13
	c12	2.64	-1.64	$94.38	$51,040.18
-2x (GLL)	c12	2.29	-1.29	$56.11	$2,371.02
2x (UGL)	c1	-1.60	2.60	$52.68	$312.98
	c2	-1.59	2.59	$50.26	$315.26
	c6	-1.55	2.55	$42.54	$332.02
	c12	-1.47	2.47	$44.60	$332.71
3x (UGLDF)	c1	-3.95	4.95	$174.51	$979.96
	c2	-3.93	4.93	$170.04	$983.41
	c6	-3.87	4.87	$152.43	$1,007.18
	c12	-3.71	4.71	$143.10	$991.18

5 Dynamic Replication

Table 11 displays the tracking error (RMSE) of the actual LETFs against the theoretical leveraged benchmark. This establishes the baseline tracking error inherent in the ETF product itself before comparing it to our replication strategies.

Table 11: LETF vs. Leveraged Benchmark RMSE

ETF	**Beta**	**RMSE (LETF vs Benchmark)**
GLD	1	$54.78
UGL	2	$1119.09
GLL	-2	$101.39
UGLDF	3	$1375.81

Table 12 summarizes the Root Mean Squared Error (RMSE) for our dynamic replication strategies across different futures maturities.

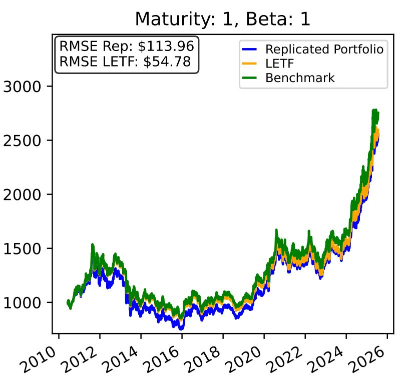
Table 12: Dynamic Replication RMSE by Futures Maturity (Replicated Portfolio vs Benchmark)

ETF Target	**Beta**	**1-Month**	**2-Month**	**6-Month**	**12-Month**
GLD	1	$113.96	$118.20	$119.39	$288.40
UGL	2	$439.63	$425.96	$395.99	$567.73
GLL	-2	$121.88	$123.16	$127.15	$105.97
UGLDF	3	$720.58	$740.78	$718.59	$529.40

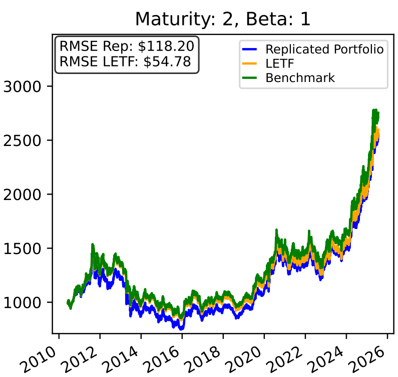
The results indicate that the the dynamic replication strategy exhibits better performance that UGL and UGLDF across all futures maturities, comparable performance to the GLL ETF, and slightly worse performance than the GLD ETF. The extremely good performance of the GLD ETF could be attributed to its high volume and AUM, enhancing its liquidity and tracking accuracy.

6 Dynamic Replication Plots

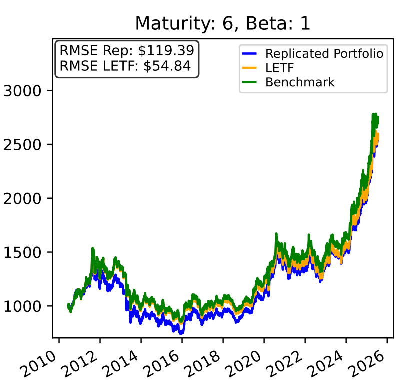
The following figures illustrate the dynamic replication performance across different futures maturities (1, 2, 6, and 12 months). Each figure displays a 2x2 grid for a specific ETF. It should be noted that the time periods shown in the plots differ between each beta, so the RMSE value are not directly comparable across the different ETFs.



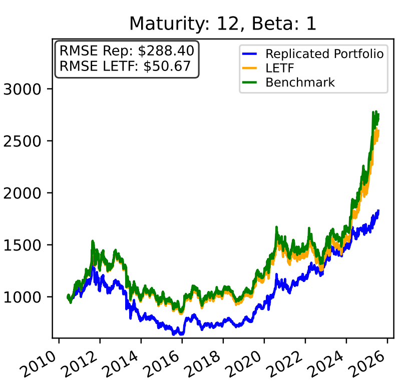
1-Month



2-Month

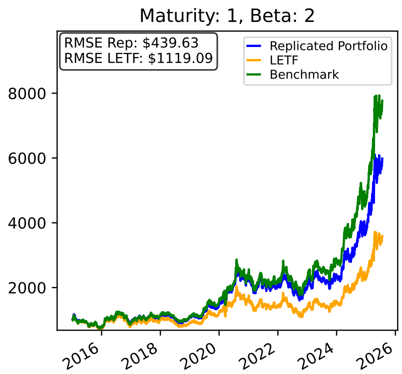


6-Month

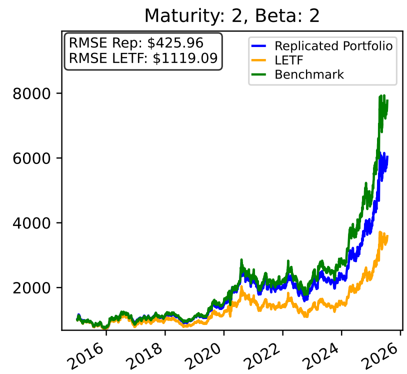


12-Month

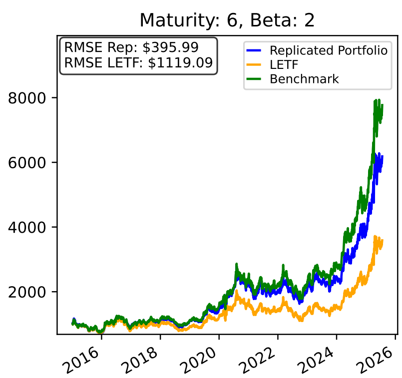
Figure 6: **GLD (1x) Replication**: The GLD ETF performs better than the replicated portfolios, but the shorter-term maturities (1m, 2m, and 6m) show reasonable performance.



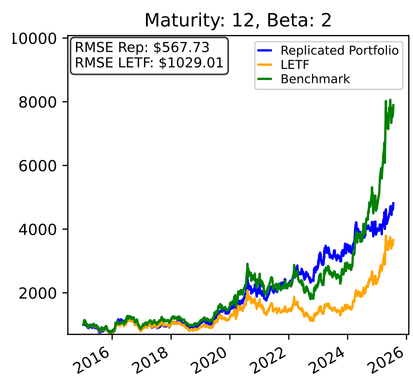
1-Month



2-Month

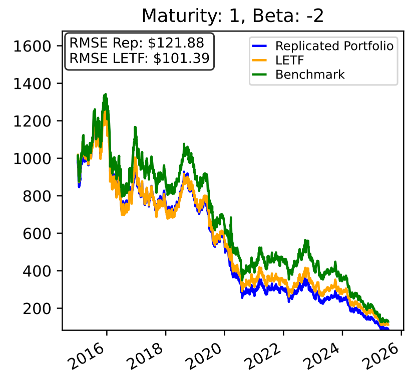


6-Month

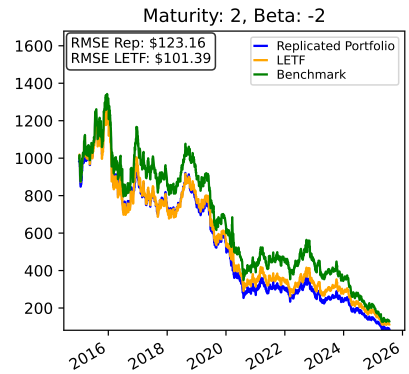


12-Month

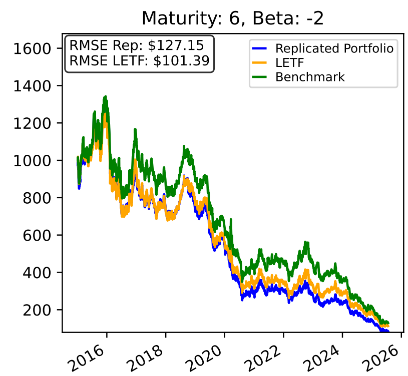
Figure 7: **UGL (2x) Replication**: The replicated portfolios exhibit significantly greater performance than the leveraged ETF. Once again, the 12m maturity performs worse than all others.



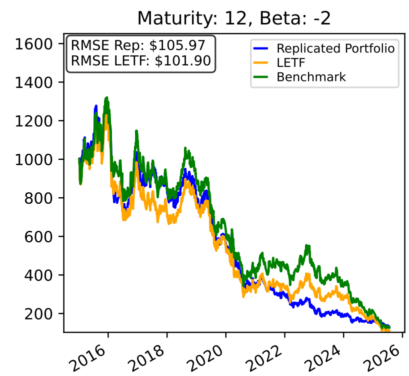
1-Month



2-Month

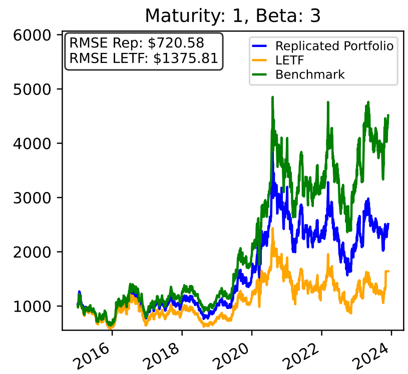


6-Month

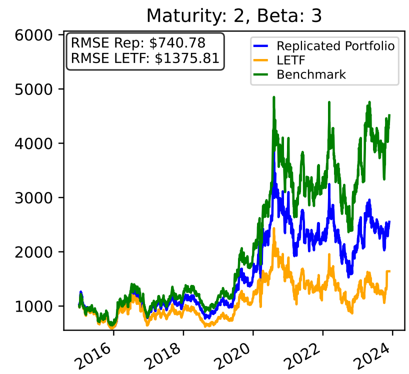


12-Month

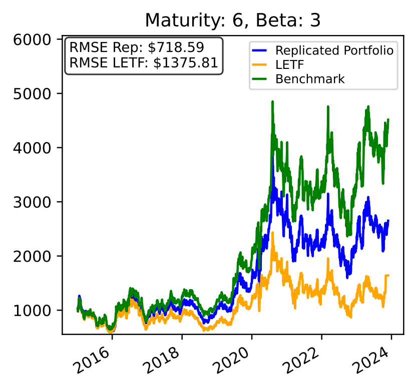
Figure 8: **GLL (-2x) Replication**: Better performance for all replicated portfolios compared to the leveraged ETF, but the 12m maturity seems less reactive to spot.



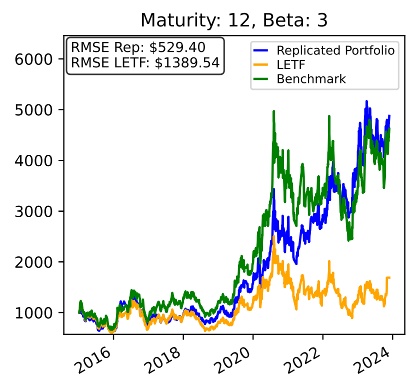
1-Month



2-Month



6-Month



12-Month

Figure 9: **UGLDF (3x) Replication**: All replicated portfolios performed better than the LETF. The higher leverage (3x) amplifies the divergence errors, particularly in the longer-dated contracts.